

SKYPORT

Group Members:

- 1) Giorgos Hadjiconstantis, hadjicog@tcd.ie, 24382117
- 2) Zuhairia Sahjabin, sahjabiz@tcd.ie, 25346560
- 3) Odysseas Leonidou, leonidoo@tcd.ie, 24383575
- 4) Teresa Domingos, domingot@tcd.ie, 17339840
- 5) Giorgos Zambas, zambasg@tcd.ie, 24382213

ABSTRACT

This software, which was made using [Processing.org](https://processing.org), allows the user to interact with a responsive data-based UI that enables said user to search and filter through a database of flights as well as display a dynamic set of charts with various data points in order to visualise the data concisely. The interface supports a user-friendly interaction, giving the user the option to exhibit a map of the United States of America with an accurate, to scale, origin and destination point of a selected flight as well as the slick and simple tutorial which guides first time users through the interface. To summarize, this is a practical application of flight searching and displaying.

LITERATURE REVIEW

Looking at how the software was constructed, key coding practices, conventional screen layout ideas, and ways to present information helped shape both code structure and the choices made during development. In order to understand key parts of programming - like organising code around objects, managing data, working out methods - official guides such as Java Documentation [4] proved vital as they provided explanations and examples that helped shape how object-oriented principles were implemented in the project. Additionally, learning sites including W3Schools [5] complemented this knowledge through accessible tutorials that guided the development of code organisation and data handling methods. Open-source platforms and forums, like Stack Overflow [6], contributed to location mapping challenges, while resources from the International Civil Aviation Organization [7], increased the accuracy of location data integration. Visuals improved once actual location details were entered into the system through sources such as OurAirports [7], making functions work more realistically. When it came to obtaining data, information about airport locations often came from applications that handle interactive flight data exploration.

Instead of static images, moving parts - like ads or looping clips [1–3, 8, 12] - shaped both appearance and how users interact. The use of dynamic visual elements enhanced user engagement by providing immediate feedback and creating a more

immersive interface. In addition, these elements improved the clarity of information presentation, as animated transitions made changes in data states more comprehensible. Alongside these, tools like sample code projects [13] gave structure, while the guides focused on motion simulation [14], which helped create chart animations with a natural flow.

Even though tools existed, the preferred approach was to tackle coding and appearance separately. This deliberate separation was chosen to create clearer boundaries between logic and interface. Furthermore, by separating communication concerns, the project addressed frequent issues such as difficulties in testing code and challenges in building responsive components. Reflecting on these choices, separating code and appearance improved maintainability, while also supporting effective adaptation as needs evolved and enabling more efficient troubleshooting.

PROJECT DETAILS

This project is an interactive flight data visualisation, the primary aim of the software is to enable users to explore, filter, and analyse flight data through a visually engaging and user-friendly interface, combining both functional data handling and dynamic visual presentation.

The system is organised into multiple interconnected screens, including a home screen, search results interface, and a data visualisation dashboard. The home screen allows one to navigate and access the system in various ways, while the search results screen enables direct interaction with the flight data. A key feature of the system is the implementation of a dynamic dropdown search, which enables users to filter all given flights based on flight preferences. This real-time filtering allows efficient data exploration. Our combined efforts led us to create a paginated system which allows the user to browse through the large flight data-base easily, ensuring smoothness and clarity in the interface.

The project makes use of object-oriented programming principles, with multiple classes such as Flight, Button, Widget, and WidgetList handling different components of the application. This modular structure allows us, as a team, to make the code much more reusable and scalable. Flight data is loaded from a CSV file containing real-world airport information, including latitude and longitude coordinates, which are used to support geographical mapping and enhance the realism of the system.

A major aspect of the project is its focus on interactive data visualisation. The system includes animated charts, such as pie charts and bubble charts, which add an intuitive element and spin to the overall UI experience. These visualisations are enhanced through the use of easing functions and spring physics techniques, which result in a much more pleasurable experience. Additionally, a map-based view allows

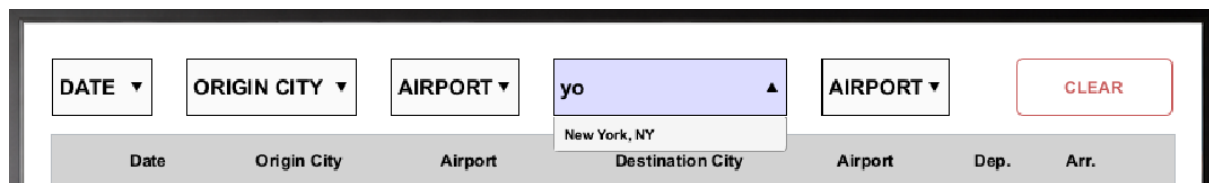
users to visualise individual flight routes (from origin to destination in a cool and interactive linear path), further improving the depth of analysis available as a result.

Event handling is a critical component of the system, managing user interactions such as mouse clicks, dropdown expansion, and page navigation as well as mouse dragging features. As a team, we dedicated a significant portion of our attention to the synchronicity and bug-free functionality of all widgets and user interactive items (especially when it came to displaying the filtered search drop-down menus). Another big challenge encountered was preventing unintended clicks on underlying buttons when dropdown menus were expanded. This issue was resolved by creating interface states that separated the event handling of certain widgets.

The decision to use Processing was based on its strong support for graphical applications and its compatibility with Java, allowing both visual and technical aspects of the project to be developed efficiently.

EXECUTION AND RESULTS

On the flights screen, we can use the search bar to filter using the drop-down buttons. In the drop-down, you can type using your keyboard to filter the available fields.



The screenshot shows a flight search interface with several dropdown menus and a search bar. The filters are: DATE, ORIGIN CITY, AIRPORT, a search bar containing 'yo' with a dropdown menu showing 'New York, NY', another AIRPORT, and a CLEAR button. Below the filters is a table header with columns: Date, Origin City, Airport, Destination City, Airport, Dep., and Arr.

After selecting filters using the drop-down buttons, the flights are searched instantly and the filtered results are shown on screen with pages of results.

DATE ▾

ORIGIN CITY ▾

AIRPORT ▾

New York, NY ▾

AIRPORT ▾

CLEAR

Date	Origin City	Airport	Destination City	Airport	Dep.	Arr.
1/1/2022	New Orleans, LA	MSY	New York, NY	JFK	1949	0004
1/1/2022	Orlando, FL	MCO	New York, NY	JFK	0000	0000
1/1/2022	Houston, TX	IAH	New York, NY	LGA	1005	1502
1/1/2022	Minneapolis, MN	MSP	New York, NY	JFK	0942	1313
1/1/2022	Palm Beach, FL	PBI	New York, NY	LGA	1700	1926
1/1/2022	Orlando, FL	MCO	New York, NY	JFK	1507	1748
1/1/2022	Seattle, WA	SEA	New York, NY	JFK	1448	2246
1/1/2022	Salt Lake City, UT	SLC	New York, NY	JFK	0132	0734
1/1/2022	Fort Lauderdale, FL	FLL	New York, NY	LGA	0000	0000
1/1/2022	Norfolk, VA	ORF	New York, NY	LGA	1749	1902
1/1/2022	Rochester, NY	ROC	New York, NY	JFK	0625	0731

1

2

3

4

5

6

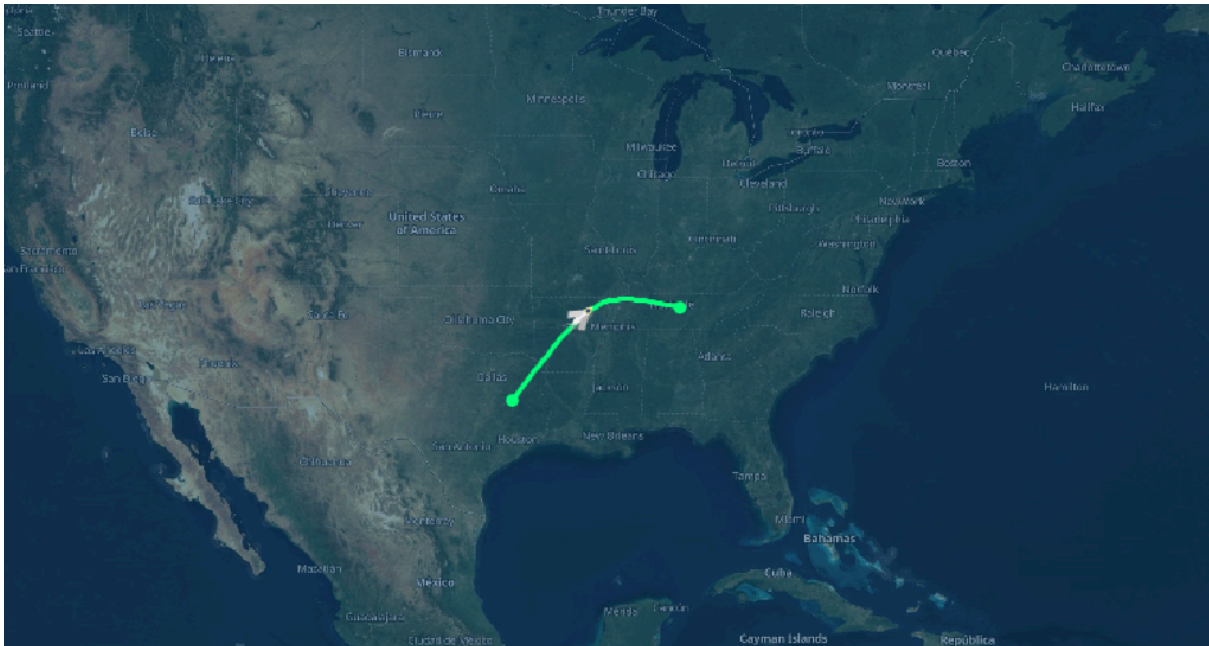
7

8

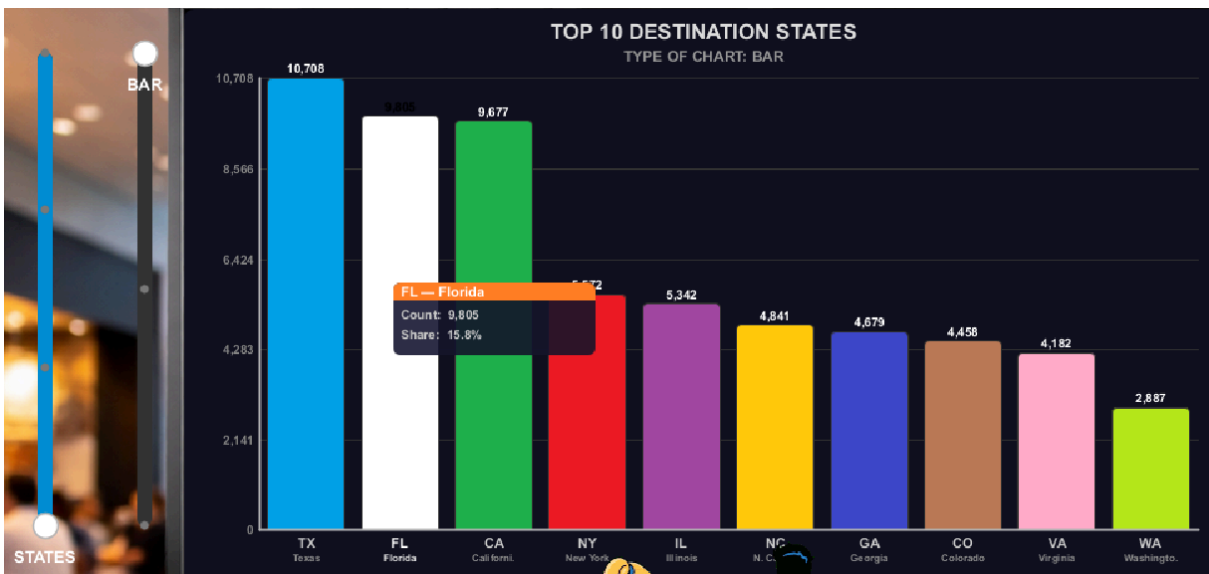
Then a specific flight can be clicked to be selected taking the user to the map screen.

1/1/2022	Baltimore, MD	BWI	Las Vegas, NV	LAS	0638	0854
1/1/2022	Houston, TX	IAH	Nashville, TN	BNA	1209	1417
1/1/2022	Oakland, CA	OAK	Santa Ana, CA	SNA	0814	0934

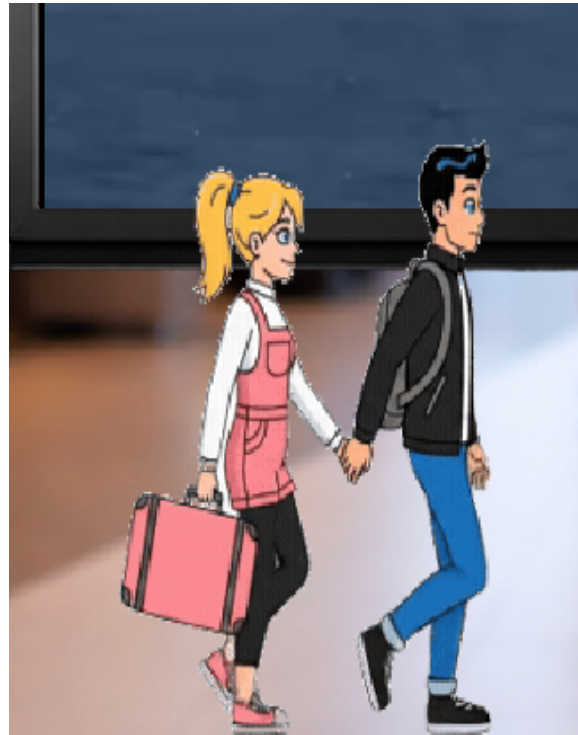
The map screen will show a plane flying from the origin airport to the destination airport and the flight path.



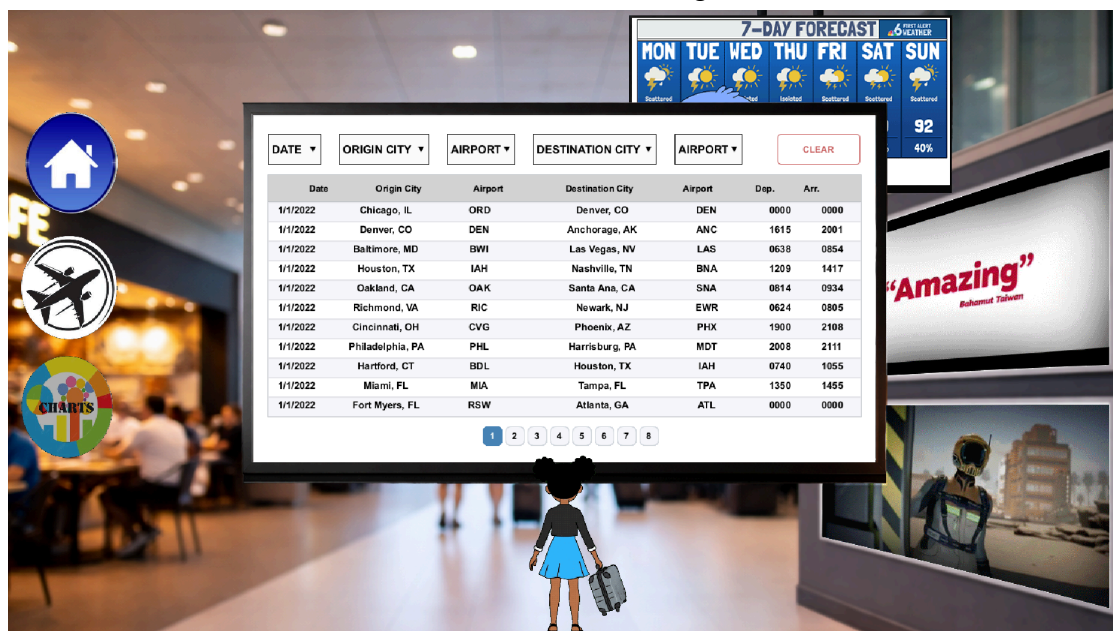
On the graph screen we have multiple interactive graphs like bar plots, pie charts and bubble charts with results showing info on top destination states, top airlines, top origins and top destinations selecting using the sliders.



On the visual side we have gifs of people walking by and looking at our virtual screen inside our airport and walking away depending on which screen is selected.



The visual style was imitating a real airport with the live flight information screens, advertisements, cafes and background noise.



CONCLUSION

In conclusion, we have successfully developed an interactive flight data visualisation program, achieving its aim of combining the handling of data with an intuitive and user-friendly interface. The software allows users to search and explore flight information through live filtering, and interactive visuals, including charts and map - based displays. Key strengths of the project is its implementation of object - oriented programming structure, its effective use of real world data, and the use of animation techniques to enhance user engagement. Throughout the development process, numerous challenges were encountered like event handling between overlapping interfaces. Overcoming these challenges required understanding of the ordering of the interfaces and state management. Overall, the project demonstrates the effective integration of Java and visual design.

REFERENCES

- [1] Steam deck advertisement by [Valve](#) found on youtube at  Steam Deck: All-in-one portable PC gaming. used in the background. (link accessed on the 25th of march 2026)
- [2] Satisfactory trailer by [Coffee Stain Studios](#) used as an advertisement in the background found here  Satisfactory - Official Console Launch Trailer . (link accessed on the 25th of march 2026)
- [3] Gif used in background for the weather found here: <https://tenor.com/view/summer-weather-climate-change-climate-heatwave-gif-8223632875427502494>. (link accessed on the 25th of march 2026)
- [4] The official Java documentation was referenced when developing our program: <https://docs.oracle.com/en/java/> (link accessed on the 11th of march 2026)

[5] W3Schools was used for documentation on java functions and methods.
<https://www.w3schools.com/java/> (link accessed on the 11th of march 2026)

[6] Stack overflow was used for referencing other peoples solutions to problems in java. <https://stackoverflow.com/questions> (link accessed on the 11th of march 2026)

[7] These websites were used to collect the airport latitude and longitude.
<https://www.iata.org/en/publications/manuals/airline-airport-location-coding-database/s/airport-location-identifier-database/>

<https://www.latlong.net/category/airports-236-19.html>

<https://ourairports.com/countries/US/airports.csv>

This is the final altered dataset that was used

<https://drive.google.com/file/d/1GhcSvHmZznxe5ExrkeM16RD9skWlp2eE/view?usp=sharing>

[Accessed Date: 15 March 2026]

[8] These links were used for the background music of the software

https://youtu.be/CN99Let_Lck?si=e1bcnAEb2lw9xGRN

https://youtu.be/245U_kF0qps?si=SQDVQa4aiKfYTEcc

<https://youtu.be/0fHezXle-h4?si=v3TF0b46cvGqTuMc>

<https://youtu.be/Huzr4MAoecc?si=bjdx7wu6A1IL0-Eg>

[Accessed Date: 24 March 2026]

[9] This site was used as reference for the homescreen button

<https://autosphereusa.com/products/husky-liner-96891-ioniq-5-weatherbeater-front-second-row-floor-liners-in-black>

[Accessed Date: 1 April 2026]

[10] The image was used as the flight screen

<https://www.vecteezy.com/vector-art/41412191-flight-icon-vector-symbol-plane-icon-isolated-on-white-background>

[Accessed Date: 1 April 2026]

[12] The image for the person walking found in GIPHY

<https://giphy.com/gifs/KSFmJd52Ue3uBbFToN>

<https://giphy.com/gifs/LSh77wodC3YPcrTlgu>

[Accessed Date: 23 March 2026]

[13] The process done for the animation of graphs

<https://processing.org/examples/easing.html>

[14] This website was used for inspiration for spring physics of bubble chart

<https://www.gorillasun.de/blog/spring-physics-and-connecting-particles-with-springs/>

[15] Map of the United States

<https://drive.google.com/file/d/13GunJfYaD3VrPzxn6Jscn4aF8sdowebO/view?usp=sharing>